

Aim - To write SQL queries for creating & using views in a database.

Theory A view in SQL is a virtual table created from one or more existing tables. It does not ~~have~~ store data physically, instead, it displays data from base tables using a query.

views help in;

- Data Security
- Simplifying complex queries
- Hiding unnecessary data
- Providing customized data access.

Advantages of view

1. Improves data security
2. Reduces query complexity
3. Provides logical data independency
4. Allows restricted access of data.

Syntax

```
CREATE view view_name AS  
SELECT column1, column2  
FROM table_name  
WHERE condition;
```


Program

```
Select Database  
USE College.DB;  
Create Student table  
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Course_ID INT,  
    SEMESTER INT,  
    Email VARCHAR(100)  
);
```

Data insertion

```
INSERT INTO Student VALUES  
(1, 'Rahul', 101, 3, 'Rahul@gmail.com'),  
(2, 'Priya', 102, 4, 'priya@gmail.com'),  
(3, 'Aman', 103, 5, 'Aman@gmail.com');
```

CREATE view

```
CREATE VIEW Student_View AS  
SELECT student_id, Name, semester  
FROM Student;
```

Display

```
SELECT * FROM Student_View;
```

Output

Student Table

student_id	name	course_id	Sem	Email
1	Rahul	101	3	rahul@gmail.com
2	Priya	102	4	priya@gmail.com
3	Amen	103	5	amen@gmail.com

Student View

Student_ID	Name	Semester
1	Rahul	3
2	Priya	4
3	Amen	5

CSE - students View

Name	Semester
Rahul	3
Amen	5

```
CREATE VIEW CSE_Students AS  
SELECT Name, Semester  
FROM Student  
WHERE course_id = 101;
```

```
SELECT * FROM CSE_students;  
DROP VIEW CSE_student
```

Result Thus, SQL Queries for creating, displaying & deleting Views were successfully implemented.

Viva Question

Q1 What is view in DBMS?

Ans A view is a virtual table created from one or more tables.

Q2 Does a view store data physically?

Ans No, it stores only if SQL query, not actual data.

Q3 What is the use of view?

Ans Views simplify queries & improve data security.

Q4 Which command is used to create a view?

Ans CREATE VIEW

Q5 Which command is used to remove a view?

Ans DROP VIEW.

Aim:- To write SQL queries for implementing Triggers in a database

Theory A trigger is a special type of stored procedure that automatically executes when a specific event occurs in a table.

Triggers are associated with:

- INSERT
- UPDATE
- DELETE

Operations on a table

Triggers help in:-

- Maintaining data integrity
- Automatic validation
- Auditing changes
- Logging activities

Types of Triggers

1. BEFORE Trigger → Executes before INSERT, UPDATE or DELETE operation

2. AFTER Trigger → Execute after INSERT, UPDATE or DELETE operation.

Syntax of Trigger

```
CREATE TRIGGER trigger-name  
BEFORE / AFTER INSERT / UPDATE / DELETE  
ON table-name  
FOR EACH ROW  
BEGIN
```

trigger statements;
END;

Program

```
USE CollegedB;  
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Marks INT );
```

```
CREATE TABLE Student_log (  
    log_ID INT AUTO_INCREMENT PRIMARY KEY,  
    Student_ID INT,  
    Action_performed VARCHAR(50) );
```

```
INSERT INTO Student VALUES  
(1, 'Rahul', 85);  
(2, 'Priya', 90);
```

Output

student-table

Student-ID	Name	Mark
1	Rahul	85
3	Aman	88

Student-log table

log-ID	Student-ID	Action-Performed
1	3	Record Inserted
2	2	Record Deleted

```
CREATE TRIGGER Student_INSERT_Trigger
AFTER INSERT
ON Student
FOR EACH ROW
INSERT INTO Student_log (Student-ID, Action-
performed)
VALUES (NEW.StudentID, 'Record inserted');
```

```
INSERT INTO Student VALUES
(3, 'Aman', 88);
```

Display

```
SELECT * FROM Student;
SELECT * FROM Student_log;
```

```
or CREATE before delete trigger
CREATE TRIGGER Student_Delete_trigger
BEFORE DELETE
ON Student
FOR EACH ROW
INSERT INTO Student_log (Student-ID, Action-
performed)
VALUES (OLD.Student-ID, 'Record Deleted');
```

```
DELETE FROM Student
WHERE Student-ID = 2;
SELECT * FROM Student_log
```


Result Thus, SQL Queries for implementing Triggers, were successfully executed in the database.

Viva Question.

Q1 What is a Trigger?

Ans A trigger is a stored procedure that automatically executes when an event occurs on a table.

Q2 What are the types of trigger?

Ans Before trigger & After trigger.

Q3 On which operations can triggers be applied?

Ans INSERT, UPDATE, DELETE.

Q4 What is the use of Triggers?

Ans Triggers help in automatic validation, logging & maintaining data integrity.

Q5 What are OLD & NEW keywords in triggers?

Ans OLD refers to old data before modification.

NEW refers to new data after modification.